

Project 2: Iris Flower Classification Model

Step 1 — Import Libraries

```
import numpy as np import pandas as pd import matplotlib.pyplot as plt import seaborn as
sns from sklearn.datasets import load_iris from sklearn.model_selection import
train_test_split from sklearn.preprocessing import StandardScaler from sklearn.neighbors
import KNeighborsClassifier from sklearn.metrics import (accuracy_score,
classification_report, confusion_matrix)
```

Step 2 — Load and Explore the Dataset

```
# Load dataset iris = load_iris() X = iris.data # Features (4 columns) y = iris.target #
Labels (0=setosa, 1=versicolor, 2=virginica) # Wrap in DataFrame for easy viewing df =
pd.DataFrame(X, columns=iris.feature_names) df['species'] = pd.Categorical.from_codes(y,
iris.target_names) print(df.head()) print(df.describe())
print(df['species'].value_counts())
```

Step 3 — Split into Train & Test Sets

```
# 80% training, 20% testing; random_state for reproducibility X_train, X_test, y_train,
y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)
print(f"Training samples: {len(X_train)}") print(f"Testing samples : {len(X_test)}")
```

Step 4 — Feature Scaling

```
scaler = StandardScaler() X_train = scaler.fit_transform(X_train) # Fit on train,
transform train X_test = scaler.transform(X_test) # Transform test (no fit!)
```

Step 5 — Train the Model (K-Nearest Neighbors)

```
# KNN with k=5 is a solid baseline for Iris model = KNeighborsClassifier(n_neighbors=5)
model.fit(X_train, y_train) print("Model training complete!")
```

Step 6 — Evaluate the Model

```
# Predictions y_pred = model.predict(X_test) # Accuracy acc = accuracy_score(y_test,
y_pred) print(f"Accuracy: {acc * 100:.2f}%") # Detailed report
print(classification_report(y_test, y_pred, target_names=iris.target_names)) # Confusion
Matrix cm = confusion_matrix(y_test, y_pred) sns.heatmap(cm, annot=True, fmt='d',
cmap='Blues', xticklabels=iris.target_names, yticklabels=iris.target_names)
plt.title('Confusion Matrix') plt.ylabel('Actual') plt.xlabel('Predicted')
plt.tight_layout() plt.savefig('confusion_matrix.png', dpi=150) plt.show()
```

Step 7 — Predict a New Sample

```
# Example: new flower measurements [sepal_len, sepal_wid, petal_len, petal_wid]
new_sample = np.array([[5.1, 3.5, 1.4, 0.2]]) new_sample_scaled =
scaler.transform(new_sample) prediction = model.predict(new_sample_scaled) species_name =
iris.target_names[prediction[0]] print(f"Predicted species: {species_name}")
```